

Fixed-Point Grasping with RoboMaster EP

Experiment 2 Group Report

SIMULATION

5 / 5

MEAN ERROR

4.358 mm

HARDWARE

5 / 5

REPOSITORY

Public

Team members

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Experiment

S16-S20 / E5

Platform

DJI RoboMaster EP arm and gripper

Report date

13 September 2026

Repository

github.com/ccx8kg5qpd-debug/robomaster-ep-fixed-point-grasping

1 Experiment Scope and Team Organization

1.1 Objective and equipment adaptation

The task is fixed-point pick-and-place without visual localization: return home, approach point A, descend, grasp, lift, move to point B, release and return home. The required acceptance target is at least four successful transfers in five consecutive trials, together with collision, joint-limit and failure checks.

Because the originally specified standalone mechArm 270 equipment was unavailable, the group used the arm and gripper mounted on a DJI RoboMaster EP chassis. The simulation uses ROS 2 Humble, Gazebo Fortress and custom kinematics/control; the physical test uses the RoboMaster Python SDK. This report treats that as a platform adaptation, not as proof of an unchanged ROS 2 hardware backend.

1.2 Division of work

Member	Area	Main responsibility
Bo Ning	Simulation	Environment, scene, models, resources and launch setup.
Junbiao Huang	Simulation	Motion program, feedback logic and repeated-cycle debugging.
Yongyue Qi	Analysis	Result/log audit, error analysis, report materials and repository integration.
Yutong Xue	Hardware	Pick/release waypoints, approach/lift height and gripper-retention checks.
Lintai Bao	Hardware	SDK communication, task sequence, chassis turn/return and exception handling.

Xue and Bao jointly performed physical setup, parameter review and integrated trials. Their scopes distinguish technical focus without reducing either member to a passive observer. All five members share the final project outcome.

1.3 Outcome summary

The recorded simulation completed five placements and verified home after all five cycles. The team confirmed five successful hardware transfers on 12 September 2026 with no observed abnormality. Numerical accuracy is available for the simulation only; the hardware result is observational.

Evidence boundary. The group did not supply measured hardware position error, force, payload mass or per-cycle duration. Those quantities are not inferred from a 5/5 observed outcome.

2 System Principle and Task Sequence

2.1 Two implementations of the same task intent

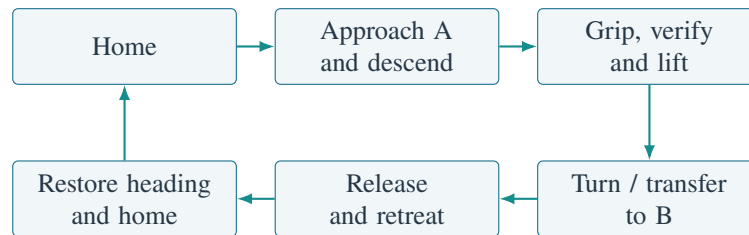


Figure 1: Shared task intent. Simulation and hardware implement the sequence through different control interfaces and cycle boundaries.

2.2 Simulation kinematics and feedback

The simulated arm is represented by two independent arm variables, r and l , with linked joint coordinates such as

$$q_{\text{arm1}} = r, \quad q_{\text{arm2}} = l - r, \quad q_{\text{endpoint}} = -l, \quad q_{\text{rod}} = l.$$

The inverse calculation minimizes fingertip-centre error in the model's x - z plane and rejects residuals above 0.002 m or solutions outside the saved joint and coupling limits. Each commanded joint transition uses the quintic blend $s(u) = 10u^3 - 15u^4 + 6u^5$ to avoid discontinuous reference velocity.

Fresh bilateral gripper contact, retained-object height and placement conditions gate later phases. Placement requires horizontal error below 23 mm, height error below 8 mm and table contact. Home acceptance separately checks heading, translation and arm error for a stable interval.

2.3 Physical control workflow

The final hardware script initializes the RoboMaster in AP connection mode and commands the arm, gripper and chassis through the Python SDK. After each lift it stops the chassis, clears old keyboard input and waits up to 30 s for a fresh human confirmation that the object is retained. Only then is chassis rotation allowed. Odd cycles turn $+32^\circ$ from A to B; even cycles turn -32° from B to A. After cycle 5 the remaining heading is restored before the arm moves through HIGH to HOME.

The confirmation is a supervised safety checkpoint, not a force or vision measurement. Cleanup pauses the gripper, commands zero wheel speed and closes the SDK connection, but it is not described as a certified emergency-stop system.

3 Simulation Development and Results

3.1 Scene, resources and execution

The scene includes a $1.6 \times 1.2 \times 0.05$ m table, a 25 mm cube and fixed points $A = (0.278, -0.001, 0.0725)$ m and $B = (0.237, 0.1453, 0.0725)$ m. The cube's simulation mass is 50 g; that setting is not a measurement of the real object. The launch file starts Gazebo, the bridge, model spawning, state feedback and the task controller. Bundled community model resources are attributed to [jeguzzi/robomaster_ros](#), archived at commit `c05a39d7f0fa`.

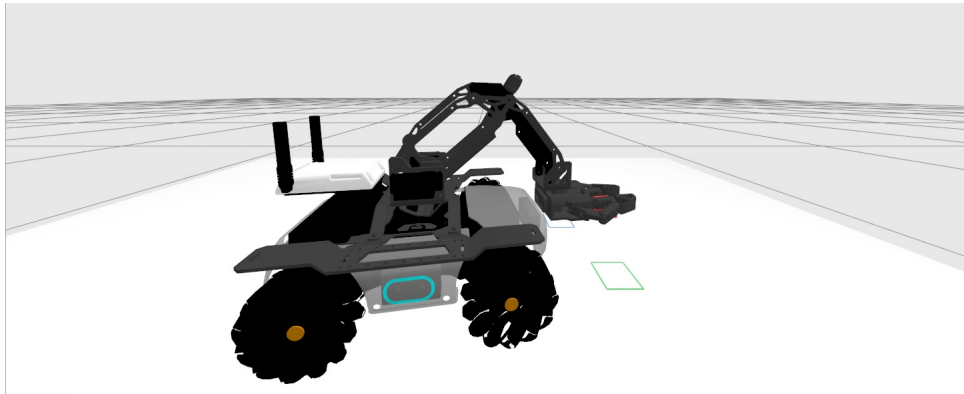


Figure 2: Recorded simulation frame during a lifted-object phase. This image is simulation evidence, not a photograph of hardware accuracy.

3.2 Five-cycle recorded result

Cycle	Error (mm)	Heading (°)	Time (s)	Placed	Home
1	2.274	0.445	67.01	Yes	Yes
2	5.604	0.329	59.72	Yes	Yes
3	4.192	0.404	60.92	Yes	Yes
4	4.577	0.459	61.09	Yes	Yes
5	5.144	0.405	61.29	Yes	Yes

The mean horizontal error was 4.358 mm and the maximum was 5.604 mm. The five cycle intervals totalled 310.03 s; the run from initial home entry to final completion lasted 318.02 s. The trajectory file contains 4475 rows over a longer span because it includes idle recording, so row span is not used as task time.

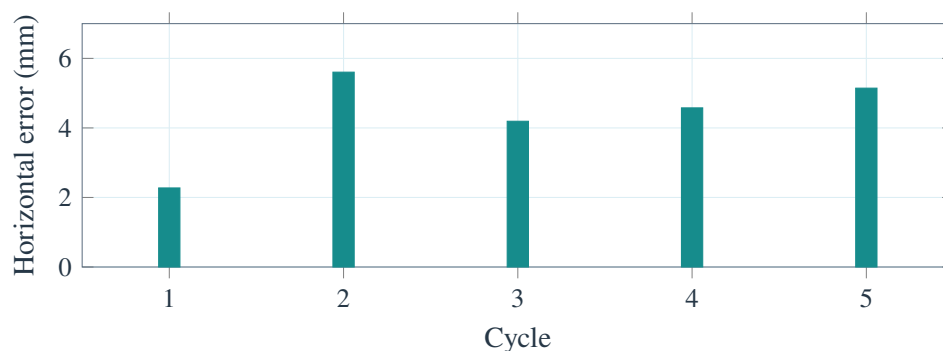


Figure 3: Horizontal placement error in the main recorded simulation run.

4 Hardware Validation and Video Evidence

4.1 Final physical parameters

Waypoint	x (mm)	y (mm)	Function
HOME	82	80	Retracted high position
HIGH	191	80	Extended high position
HOVER	191	-1	Approach and retained-object height
GRASP	191	-51	Pick and release level

The arm descends from HOVER to GRASP, closes the gripper at SDK power 100, lifts 50 mm and waits for human confirmation. The final hardware workflow alternates $A \rightarrow B$ and $B \rightarrow A$ because it does not automatically replenish the object. This differs from the simulation, where a separate refill node resets the cube to A only between cycles 1–4 after home verification.

4.2 Observed five-cycle result

Cycle	Direction	Lift confirmed	Transfer	Abnormality
1	$A \rightarrow B$	Yes	Successful	None observed
2	$B \rightarrow A$	Yes	Successful	None observed
3	$A \rightarrow B$	Yes	Successful	None observed
4	$B \rightarrow A$	Yes	Successful	None observed
5	$A \rightarrow B$	Yes	Successful	None observed

This table records the team's confirmed outcome, not a sensor-generated hardware test log. No hardware placement-error or timing series is available.

4.3 Real-device video delivered separately

The real-device video is a separate submission artifact and is not embedded in this PDF. Visual review confirms that it shows the RoboMaster EP arm, gripper, target cubes and operator terminal.

Field	Recorded value
Archive filename	RoboMaster_EP_Hardware_Demonstration.mp4
Original filename	b635ef9bacf591f62e65e10741d55908.mp4
Duration and frame	74.87 s; 1280×720 at 29.97 fps
Encoding	HEVC Main video; AAC LC stereo audio
Size	8,024,795 bytes
SHA-256	614a8eb55d8883cce4494496a199c828bb679b693f58a325319570a89fad6a6e

5 Problems, Solutions and Evidence Limits

5.1 Model resources and reference frames

Early model problems included unresolved mesh paths and inconsistent end-effector references. The final archive bundles the model dependency and provides `prepare_paths.py` to construct a runtime copy with local mesh paths. The kinematic calculation uses the forward ends of gripper link 4 rather than a rear linkage. This separates a resource-resolution failure from a control failure.

5.2 Unstable lift and false closure

A gripper-close command is not evidence that an object is held. The final simulation combines bilateral fresh contact, a small closure preload, a hold interval and an object-height check before transfer. The physical script uses a 50 mm lift followed by a fresh human confirmation. Neither mechanism proves gripping force, but each prevents a simple command-completion flag from being treated as a successful grasp.

5.3 Repeated-cycle state and heading restoration

Earlier repeated sequences could drift or stop after one cycle. The final simulation explicitly returns heading and home, records five placement checks, five home checks and four refill acknowledgements, and ends after cycle 5 without an extra refill. The physical script alternates directions and compensates the remaining heading after cycle 5.

5.4 Abnormal-case evidence

The archived notes record eight passing offline tests across model and five-cycle checks. An unreachable target such as (10, 10, 0.6) is rejected by the kinematic solver with a `ValueError`; joint and coupling limits are checked before motion. Runtime state-feedback and phase-timeout guards also exist in the controller. These are useful safeguards, but no separate system-level fault-injection run or individual safety-check record was supplied.

5.5 Deviations from the handout

Handout expectation	Evidence status
mechArm 270 hardware	Replaced by RoboMaster EP because of equipment availability.
Same ROS 2 interface for hardware	Not demonstrated; hardware uses the RoboMaster Python SDK.
Parameter-only migration	Not demonstrated across the different control backends.
Each member's individual safety check	No individual check record supplied.
At least 4/5 transfers	Met in both recorded simulation and team-observed hardware outcome.

These deviations are reported rather than hidden. They should be considered when interpreting the result and when the instructor evaluates platform equivalence.

6 Repository and Version Traceability

The public shared repository is github.com/ccx8kg5qpd-debug/robomaster-ep-fixed-point-grasping. The experiment was completed before repository assembly, so the history below records the real review, organization and publication sequence on 13 September 2026; it is not presented as a fabricated history of the experimental development itself.

ccx8kg5qpd-debug / robomaster-ep-fixed-point-grasping			main
Public GitHub repository commit evidence captured after hardware-video upload			
10 genuine archive and publication commits 13 September 2026 PDT			
● media(hardware): add real-device demonstration video	Yongyue Qi committed on 13 Sep 2026	21:14 PDT	4306259
● docs(reports): remove placeholder-like wording	Yongyue Qi committed on 13 Sep 2026	21:11 PDT	dc24e3
● docs(reports): align public repository wording	Yongyue Qi committed on 13 Sep 2026	21:01 PDT	9f87cc8
● docs(reports): publish five individual LaTeX reports	Yongyue Qi committed on 13 Sep 2026	20:49 PDT	29309a1
● docs: capture published commit history for reports	Yongyue Qi committed on 13 Sep 2026	20:37 PDT	55e823a
● media: add complete five-cycle simulation recording	Yongyue Qi committed on 13 Sep 2026	20:32 PDT	6cd8938
● data: add verified run records and course traceability	Yongyue Qi committed on 13 Sep 2026	20:32 PDT	35135a2
● feat(hardware): add final five-cycle SDK sequence	Yongyue Qi committed on 13 Sep 2026	20:32 PDT	ed65fb2
● feat(simulation): archive controller scene and model resources	Yongyue Qi committed on 13 Sep 2026	20:31 PDT	d37ae67
● docs: establish shared experiment archive	Yongyue Qi committed on 13 Sep 2026	20:29 PDT	80fff9d
Source: local main-branch Git log after every listed commit was pushed to the public GitHub repository.			
The experiment predates repository assembly; the timeline documents later review and publication, not fabricated development history.			

Figure 4: Ten repository commits visible after the real-device video was uploaded and before this group report was added.

The individual reports contain an earlier five-commit snapshot that was accurate at their capture time. Later commits add the report set, update public-repository wording and add the real-device video. Both images therefore document genuine states of the same history at different publication stages.

7 Submission Manifest and Conclusion

7.1 Materials required by the experiment handout

Required material	Status	Location in group package
ROS 2 program, launch and parameters	Included	01_simulation/
Simulation model and point/height setup	Included	01_simulation/model_dependency/ and configuration files
Simulation demonstration video	Included	03_videos/simulation/
Real-device demonstration video	Included	03_videos/hardware/
Five-cycle result records	Included	04_results/20260913_153855_682865/
Abnormal-test record	Included with limits	04_results/validation_record.md and report Section 5.4
Brief group experiment report	Included	05_group_report/
Five individual reports	Included	06_individual_reports/
GitHub link and commit evidence	Included	Report Section 6 and repository README

7.2 Reproducibility notes

The simulation targets ROS 2 Humble and Gazebo Fortress on the team's Jetson Linux environment. The archived startup wrapper contains a workspace path that must be adapted on another machine. The model dependency is bundled with source attribution. Saved `results.json`, `events.jsonl` and `trajectory.csv` support independent checking of the reported simulation numbers without rerunning Gazebo.

7.3 Conclusion

The available evidence supports five successful simulation cycles with home verification and five team-confirmed physical transfers without observed abnormality. The project demonstrates a complete fixed-point sequence on the available RoboMaster EP platform. Its strongest quantitative evidence is the saved simulation record; the physical result is supported by the team-observed outcome and a separate real-device video, not by instrumented precision data.

7.4 Sources

- [1] Course handout, *Robot Integration Group Project I: Fixed-Point Robotic-Arm Grasping*, S16–S20 / E5, including the submission-material list.
- [2] Public team repository, [robomaster-ep-fixed-point-grasping](#), main branch, 13 September 2026.
- [3] Primary simulation record 20260913_153855_682865: results, events and trajectory; comparison record 20260913_151835_077998.
- [4] Real-device video metadata and SHA-256 recorded in `media/hardware/README.md`.
- [5] Community model source, [jeguzzi/robomaster_ros](#), archived commit c05a39d7f0fa.